

Bond-by-bond loss-absorption results

The programme produced 12 positive, 13 negative and 1 unresolved results for 26 bond series. These are qualified readings of dated offering documents. Preferred-share securities are excluded. A positive identifies principal write-down or compulsory common-share conversion, including disclosed statutory bail-in. A negative requires affirmative debt/repayment terms and no unresolved potentially qualifying candidate in the declared source set. Ordinary creditor-approved restructuring is treated separately.

1. Tesco treasury 3.500% senior notes due 2033

ISIN / identity: XS3201918409

Result: No — non loss absorption

No within the examined offering terms: the debt provides for cash principal repayment ([tesco-2025-final](#), PDF pp.2-3); the declared operative document set has been examined and no applicable principal write-down or compulsory common-share conversion was identified.

2. Veolia 3.690% senior notes due 2031

ISIN / identity: FR0014017P12

Result: No — non loss absorption

No within the examined offering terms: the debt provides for cash principal repayment ([veolia-2026-base](#), PDF p.79); the declared operative document set has been examined and no applicable principal write-down or compulsory common-share conversion was identified.

3. Veolia 4.122% senior notes due 2036

ISIN / identity: FR0014017P04

Result: No — non loss absorption

No within the examined offering terms: the debt provides for cash principal repayment ([veolia-2026-base](#), PDF p.79); the declared operative document set has been examined and no applicable principal write-down or compulsory common-share conversion was identified.

4. Deutsche Bank 6.75% perpetual AT1, December 2025

ISIN / identity: DE000A460DG7

Result: Yes — loss absorption

Yes: the principal can be reduced or cancelled under the contractual terms ([deutsche-at1-2025](#), PDF p.52). The applicable trigger conditions are stated in the cited clause; the power need not have been exercised.

5. Sogécap 6.25% perpetual restricted tier 1, July 2025

ISIN / identity: FR00140112W8

Result: Yes — loss absorption

Yes: the principal can be reduced or cancelled under the contractual terms ([sogecap-rt1-2025](#), PDF p.72). The applicable trigger conditions are stated in the cited clause; the power need not have been exercised.

6. UBS 5.600% SGD perpetual AT1 (June issue and July tap)

ISIN / identity: CH1357852636

Result: Yes — loss absorption

Yes: the selected bond can be compulsorily converted into common/ordinary shares ([ubs-sgd-at1-2024-final-published](#), PDF pp.28-29). The conversion power need not have been exercised.

7. Barclays 7.625% perpetual AT1, first reset 2035

ISIN / identity: US06738EDC66

Result: Yes — loss absorption

Yes: the selected bond can be compulsorily converted into common/ordinary shares ([barclays-at1-2025](#), PDF p.57). The conversion power need not have been exercised.

8. HSBC 6.750% perpetual AT1, first reset 2031

ISIN / identity: hsbc-at1-2026-2031

Result: Yes — loss absorption

Yes: the selected bond can be compulsorily converted into common/ordinary shares ([hsbc-at1-2026-specific](#), PDF p.66). The conversion power need not have been exercised.

9. HSBC 7.000% perpetual AT1, first reset 2036

ISIN / identity: hsbc-at1-2026-2036

Result: Yes — loss absorption

Yes: the selected bond can be compulsorily converted into common/ordinary shares ([hsbc-at1-2026-specific](#), PDF p.66). The conversion power need not have been exercised.

10. Standard Chartered 7.625% perpetual AT1, January 2025

ISIN / identity: US853254DF47, USG84228GP72

Result: Yes — loss absorption

Yes: the principal can be reduced or cancelled under the contractual terms ([standard-chartered-at1-2025-executed-deed](#), PDF pp.42-43). The applicable trigger conditions are stated in the cited clause; the power need not have been exercised.

11. Apple Inc. 4.000% Notes due 2028

ISIN / identity: apple-senior-2025.1

Result: No — non loss absorption

No within the examined offering terms: the debt provides for cash principal repayment ([apple-senior-2025](#), PDF p.16); the declared operative document set has been examined and no applicable principal write-down or compulsory common-share conversion was identified.

12. Apple Inc. 4.200% Notes due 2030

ISIN / identity: apple-senior-2025.2

Result: No — non loss absorption

No within the examined offering terms: the debt provides for cash principal repayment ([apple-senior-2025](#), PDF p.16); the declared operative document set has been examined and no applicable principal write-down or compulsory common-share conversion was identified.

13. Apple Inc. 4.500% Notes due 2032

ISIN / identity: apple-senior-2025.3

Result: No — non loss absorption

No within the examined offering terms: the debt provides for cash principal repayment ([apple-senior-2025](#), PDF p.16); the declared operative document set has been examined and no applicable principal write-down or compulsory common-share conversion was identified.

14. Apple Inc. 4.750% Notes due 2035

ISIN / identity: apple-senior-2025.4

Result: No — non loss absorption

No within the examined offering terms: the debt provides for cash principal repayment ([apple-senior-2025](#), PDF p.16); the declared operative document set has been examined and no applicable principal write-down or compulsory common-share conversion was identified.

15. Duke Energy Corporation 5.45% Senior Notes due 2034

ISIN / identity: duke-senior-2024.1

Result: No — non loss absorption

No within the examined offering terms: the debt provides for cash principal repayment ([duke-senior-2024](#), PDF p.15); the declared operative document set has been examined and no applicable principal write-down or compulsory common-share conversion was identified.

16. Duke Energy Corporation 5.80% Senior Notes due 2054

ISIN / identity: duke-senior-2024.2

Result: No — non loss absorption

No within the examined offering terms: the debt provides for cash principal repayment ([duke-senior-2024](#), PDF p.15); the declared operative document set has been examined and no applicable principal write-down or compulsory common-share conversion was identified.

17. The Southern Company Series 2025A 6.50% Junior Subordinated Notes due 2085

ISIN / identity: southern-junior-2025a.1

Result: No — non loss absorption

No within the examined offering terms: the debt provides for cash principal repayment ([southern-junior-2025a](#), PDF p.10); the declared operative document set has been examined and no applicable principal write-down or compulsory common-share conversion was identified.

18. Duke Energy Corporation 6.45% Fixed-to-Fixed Reset Rate Junior Subordinated Debentures due 2054

ISIN / identity: duke-junior-2024.1

Result: No — non loss absorption

No within the examined offering terms: the debt provides for cash principal repayment ([duke-junior-2024](#), PDF p.12); the declared operative document set has been examined and no applicable principal write-down or compulsory common-share conversion was identified.

19. NatWest Markets 4.789% senior notes due 2028

ISIN / identity: USG6382RGD47, US63906YAM03

Result: Yes — loss absorption

Yes: the principal can be reduced or cancelled under a disclosed statutory resolution power ([nwm-2025-base](#), PDF p.69). The applicable trigger conditions are stated in the cited clause; the power need not have been exercised.

20. BPCE USD SOFR + 1.03% senior preferred notes due 2030

ISIN / identity: FR0014014YQ1

Result: Yes — loss absorption

Yes: the principal can be reduced or cancelled under a disclosed statutory resolution power ([bpce-2025-base](#), PDF p.150). The applicable trigger conditions are stated in the cited clause; the power need not have been exercised.

21. CBA EUR ESTR + 0.26% senior notes due October 2026

ISIN / identity: XS3206613666

Result: No — non loss absorption

No within the examined offering terms: the debt provides for cash principal repayment ([cba-2025-6700-final](#), PDF p.3); the declared operative document set has been examined and no applicable principal write-down or compulsory common-share conversion was identified.

22. Compass Group 3.500% senior notes due 15 January 2035

ISIN / identity: XS3232968985

Result: No — non loss absorption

No within the examined offering terms: the debt provides for cash principal repayment ([compass-2035-final](#), PDF p.2); the declared operative document set has been examined and no applicable principal write-down or compulsory common-share conversion was identified.

23. ING Groep 7.000% perpetual AT1, first reset November 2032

ISIN / identity: US456837BT90

Result: Yes — loss absorption

Yes: the selected bond can be compulsorily converted into common/ordinary shares ([ing-at1-us456837bt90](#), PDF p.36). The conversion power need not have been exercised.

24. Shell International Finance 1.000% senior notes due 10 December 2030

ISIN / identity: XS2228291279

Result: Unresolved

No supported yes/no answer: Affirmative principal repayment terms have not been resolved

25. Danske Bank DKK1.25bn perpetual AT1, Series 4, August 2026

ISIN / identity: DK0030577200

Result: Yes — loss absorption

Yes: the selected bond can be compulsorily converted into common/ordinary shares ([danske-at1-2026-base](#), PDF p.39). The conversion power need not have been exercised.

26. Lloyds 6.625% perpetual AT1, first reset September 2035

ISIN / identity: US539439BF59

Result: Yes — loss absorption

Yes: the selected bond can be compulsorily converted into common/ordinary shares ([lloyds-at1-2025-yahoo-filing](#), HTML text unit 1). The conversion power need not have been exercised.

What was checked

447 prospectus tests passed, including 305 tests for this classifier. The independent specification checked all 256 known/unknown/conflict input states; two SMT equivalence obligations passed and four deliberately wrong formal rules were detected. RuleIR/Java and Catala executed the same 26 bond input sets, plus 24 formal challenge cases in each of 4 runs, for 244 native executions. Kernel-checked lowering and source integrity establish conditional software properties, not legal interpretation.

Every one of the 6,822 emitted evidence records was rebound to its preserved source text. Six faults involving actual PDFs/text were rejected or left unresolved. Version 1 failed an edition-substitution challenge and was repaired; Compass and ING became development cases. The new frozen version processed these fresh cases: shell-senior-2030 (unresolved), danske-dkk-at1-2026 (yes). No source-answer labels were used.

Qualifications and remaining work

The binary result is the existence of the requested prospectus feature. Positive rows can still contain unresolved subsidiary clauses: a sufficient witness establishes the feature without establishing every possible mechanism or its limits. The full JSON retains those open clauses. A negative is conditional on the declared document selection and bounded English analysis; it is not a proof that all incorporated contracts, later amendments and applicable law have been found. The system cannot independently prove unrestricted English meaning or correctness for every unknown future document. It makes that qualification explicitly and uses missing-evidence states rather than inventing certainty.

Standard Chartered uses an issuer-published executed trust deed because the retained circular carries subject-to-completion language. Deutsche Bank's German terms prevail; the English translation has not been independently proved equivalent. Lloyds uses a retained final filing mirror: official issuer-PDF acquisition failed, and byte identity with EDGAR is not established. Some US/HSBC series use filing/series identities because their ISINs were not independently resolved. Issuer names, domicile and rank are metadata, never classification rules.

The fresh Shell example remains unresolved because the current reader does not bind its stated cash redemption amount to the calculation unit. That is an identified implementation limit, not evidence that the bond absorbs capital losses. The fresh-family challenge is therefore incomplete: one positive transfer and one safe abstention, with no supported legal-accuracy score.

Next work: strengthen semantic subject/condition resolution, close incorporated-document and amendment dependencies, validate authoritative language alignment, and run further frozen document-family challenges. Treat future source or law changes as reasons to re-acquire and re-evaluate. A negative here does not mean low investment risk or transaction permission.

[Machine-readable results](#) · [CSV](#) · [verification](#) · [reviewed execution plan](#) · [source freeze](#)

Correctness review — 29 September 2026

The source review supports the reported feature distinction, with qualified negative readings and one unresolved case. It does not establish that the reader interprets every prospectus correctly. Reading a repayment clause cannot, by itself, prove that a different clause never permits a write-down. The negative decisions still depend on the declared document selection and the bounded clause recogniser. Test and quotation counts do not measure legal accuracy.

Two supporting explanations needed correction. Standard Chartered's previous summary cited a Newco reorganisation provision on PDF page 55. That provision alone does not establish the mandatory trigger. Condition 7(a)(i) on PDF page 41 explicitly provides automatic conversion into ordinary shares after a Conversion Trigger Event; Condition 7(a)(vi), spanning PDF pages 42–43, also provides a full principal write-down after a Non-Qualifying Relevant Event followed by a Conversion Trigger Event. The revised generated summary cites the latter clause, already present in the retained evidence. Its positive answer remains supported. Duke's 2054 senior issue previously cited the 2034 series' redemption clause. The revised summary selects the clause for the 2054 Notes on PDF page 15. The negative answer remains qualified by the same document-completeness limits.

These are real explanation defects despite the passing tests. The summary selection was repaired and tested. The frozen reader still contains the overbroad source candidates; they remain visible in JSON and require a future semantic repair. The presentation correction is not evidence that these underlying interpretation defects have been eliminated.

For the requested corporate examples, Tesco and Compass specify repayment at 100% of nominal amount. Veolia's two final terms specify €100,000 redemption per €100,000 note (PDF page 5), consistent with the base prospectus's nominal repayment provision. No qualifying principal-loss mechanism was identified in their declared offering terms. NatWest and BPCE differ: their cited terms explicitly allow statutory bail-in to reduce principal, so their senior status does not make the feature negative. CBA's negative depends in particular on excluding ordinary creditor-approved restructuring: its Condition 13 permits collective modifications that can bind dissenting holders. That exclusion is an explicit working assumption, not a proof that principal can never be lost.

Shell remains unresolved in the programme. Its final terms give a £1,000 calculation amount on PDF page 2 and £1,000 final redemption per calculation amount on page 3. Those amounts agree, but the frozen reader does not compose the two fields. The abstention reports an implementation limit; it supplies no evidence of a loss-absorption feature. No answer has been forced merely to meet an expected negative.

This feature test is separate from the SFC's overall product-complexity and selling rules. The [SFC's examples of complex products](#) also include perpetual and subordinated bonds and other special features. Thus a junior bond can receive a negative result here and still be complex under those rules. The [SFC's 2018 circular](#) describes capital-ratio and government/regulatory triggers for non-viability loss absorption. This report does not decide transaction eligibility.

The independent formal checks establish the decision and translation logic under the supplied premises. Source completeness, unrestricted language interpretation and performance on all future instruments remain unproved. No human quality labels or source-answer accuracy score were used in this review.